

Consumption-Saving

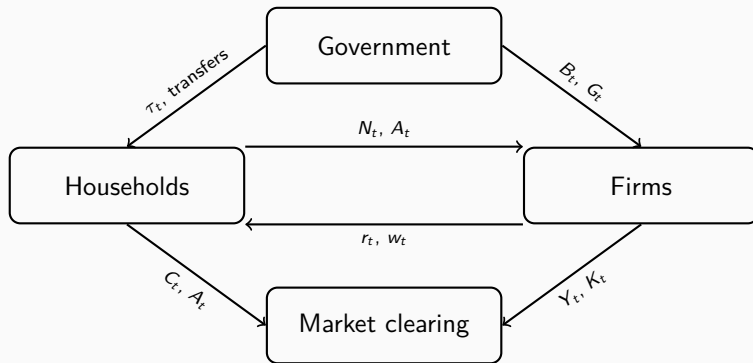
Adv. Macro: Heterogeneous Agent Models

Raphaël Huleux

2026

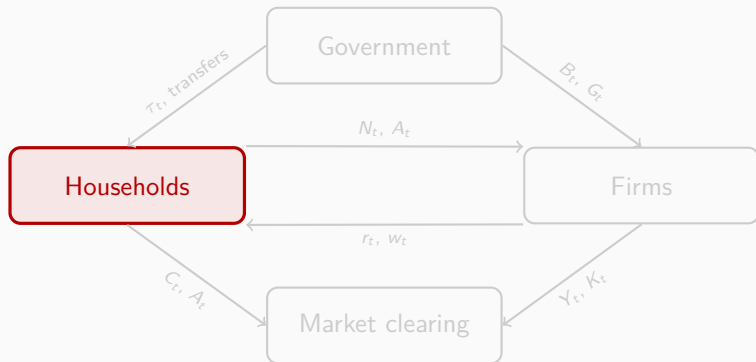
Introduction

A model is a set of blocks



Every block is a set of equations mapping inputs into outputs.

We open the household block



This course focuses on the household block.

The question: *when does the distribution matter for aggregates?*

- **Review consumption-saving theory**

1. The permanent income hypothesis, and what it gets wrong
2. The buffer-stock model: risk and borrowing constraints

Goals for today

- **Review consumption-saving theory**
 1. The permanent income hypothesis, and what it gets wrong
 2. The buffer-stock model: risk and borrowing constraints
- **Solve the household problem in partial equilibrium**
 1. Basics of dynamic programming
 2. Value function iteration (VFI)
 3. The endogenous grid method (EGM)

Goals for today

- **Review consumption-saving theory**
 1. The permanent income hypothesis, and what it gets wrong
 2. The buffer-stock model: risk and borrowing constraints
- **Solve the household problem in partial equilibrium**
 1. Basics of dynamic programming
 2. Value function iteration (VFI)
 3. The endogenous grid method (EGM)
- **Simulate the distribution and aggregate**
 1. Monte Carlo
 2. The histogram method (Young, 2010)

- **Review consumption-saving theory**
 1. The permanent income hypothesis, and what it gets wrong
 2. The buffer-stock model: risk and borrowing constraints
- **Solve the household problem in partial equilibrium**
 1. Basics of dynamic programming
 2. Value function iteration (VFI)
 3. The endogenous grid method (EGM)
- **Simulate the distribution and aggregate**
 1. Monte Carlo
 2. The histogram method (Young, 2010)
- **Note:** partial equilibrium for now, prices are *fixed*.
They become equilibrium objects in the next lecture.

- **Generations of models:**
 1. **Permanent income hypothesis (PIH)** (Friedman, 1957)
or life-cycle model (Modigliani and Brumberg, 1954)
 2. **Buffer-stock consumption model**
Deaton (1991, 1992); Carroll (1992, 1997, 2019)
 3. **Multiple-asset buffer-stock consumption models**
e.g. Kaplan and Violante (2014); Harmenberg and Öberg (2021)

Book: **The Economics of Consumption**, Jappelli and Pistaferri (2017)

- **Generations of models:**
 1. **Permanent income hypothesis (PIH)** (Friedman, 1957)
or life-cycle model (Modigliani and Brumberg, 1954)
 2. **Buffer-stock consumption model**
Deaton (1991, 1992); Carroll (1992, 1997, 2019)
 3. **Multiple-asset buffer-stock consumption models**
e.g. Kaplan and Violante (2014); Harmenberg and Öberg (2021)
- **Consumption-and-saving over the life-cycle dynamic**
e.g. Gourinchas and Parker (2002); Druedahl and Martinello (2022)

Book: **The Economics of Consumption**, Jappelli and Pistaferri (2017)

- **Generations of models:**
 1. **Permanent income hypothesis (PIH)** (Friedman, 1957)
or life-cycle model (Modigliani and Brumberg, 1954)
 2. **Buffer-stock consumption model**
Deaton (1991, 1992); Carroll (1992, 1997, 2019)
 3. **Multiple-asset buffer-stock consumption models**
e.g. Kaplan and Violante (2014); Harmenberg and Öberg (2021)
- **Consumption-and-saving over the life-cycle dynamic**
e.g. Gourinchas and Parker (2002); Druedahl and Martinello (2022)
- **Empirical MPCs and income risk**
e.g. Fagereng et al. (2021); Guvenen et al. (2021)

Book: **The Economics of Consumption**, Jappelli and Pistaferri (2017)

1. Introduction
2. PIH
3. Dynamic programming
4. VFI
5. Buffer-stock
6. EGM
7. Simulation and aggregation
8. Summary
9. Misc

PIH



- **Keynesian consumption function:** $C_t = a + bY_t$
Consumption responds to *current* income. Multiplier is $\frac{1}{1-b}$.
But it lacks micro-foundations!

- **Keynesian consumption function:** $C_t = a + bY_t$
Consumption responds to *current* income. Multiplier is $\frac{1}{1-b}$.
But it lacks micro-foundations!
- **Permanent income hypothesis** (Friedman, 1957): Consumption responds to the *present value* of income.
Current income matters only through it.

- **Keynesian consumption function:** $C_t = a + bY_t$
Consumption responds to *current* income. Multiplier is $\frac{1}{1-b}$.
But it lacks micro-foundations!
- **Permanent income hypothesis** (Friedman, 1957): Consumption responds to the *present value* of income.
Current income matters only through it.
- But: PIH implies too low MPCs. We will see how to fix it

$$v_0 = \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t u(c_t)$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$a_{T-1} \geq 0, \quad a_{-1} \text{ given}$$

- **Variables:**

Consumption: c_t

Productivity: z_t

End-of-period savings: a_t (*no debt at death*)

$$v_0 = \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t u(c_t)$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$a_{T-1} \geq 0, \quad a_{-1} \text{ given}$$

- **Variables:**

Consumption: c_t

Productivity: z_t

End-of-period savings: a_t (*no debt at death*)

- **Parameters:**

Discount factor: β

Wage: w

Interest rate: r (define $R \equiv 1 + r$ as interest factor)

- **Only the terminal condition really constrains:** without $a_{T-1} \geq 0$ the household would just borrow more and more

- **Only the terminal condition really constrains:** without $a_{T-1} \geq 0$ the household would just borrow more and more
- **Substitution** implies *Intertemporal Budget Constraint* (IBC)

$$\begin{aligned}
 a_{T-1} &= Ra_{T-2} + wz_{T-1} - c_{T-1} \\
 &= R(Ra_{T-3} + wz_{T-2} - c_{T-2}) + wz_{T-1} - c_{T-1} \\
 &= R^T a_{-1} + \sum_{t=0}^{T-1} R^{T-1-t} (wz_t - c_t)
 \end{aligned}$$

- **Only the terminal condition really constrains:** without $a_{T-1} \geq 0$ the household would just borrow more and more
- **Substitution** implies *Intertemporal Budget Constraint* (IBC)

$$\begin{aligned}
 a_{T-1} &= Ra_{T-2} + wz_{T-1} - c_{T-1} \\
 &= R(Ra_{T-3} + wz_{T-2} - c_{T-2}) + wz_{T-1} - c_{T-1} \\
 &= R^T a_{-1} + \sum_{t=0}^{T-1} R^{T-1-t} (wz_t - c_t)
 \end{aligned}$$

- Use **terminal condition** $a_{T-1} = 0$ (equality due to utility max.)

$$R^{-(T-1)} a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t} c_t = s_0 + h_0$$

where $s_0 \equiv Ra_{-1}$ (after-interest assets)

and $h_0 \equiv \sum_{t=0}^{T-1} R^{-t} wz_t$ (human capital)

$$\mathcal{L} = \sum_{t=0}^{T-1} \beta^t u(c_t) + \lambda \left[s_0 + h_0 - \sum_{t=0}^{T-1} R^{-t} c_t \right]$$

■ It is a *static* problem:

1. **Information:** s_0 and h_0 are known at $t = 0$
2. **Target:** Discounted utility, $\sum_{t=0}^{T-1} \beta^t u(c_t)$
3. **Behavior:** Choose $c_0, c_1, \dots, c_{T-1}$ *simultaneously*
4. **Solution:** Sequence of consumption *choices* $c_0^*, c_1^*, \dots, c_{T-1}^*$

$$\mathcal{L} = \sum_{t=0}^{T-1} \beta^t u(c_t) + \lambda \left[s_0 + h_0 - \sum_{t=0}^{T-1} R^{-t} c_t \right]$$

- It is a *static* problem:

1. **Information:** s_0 and h_0 are known at $t = 0$
2. **Target:** Discounted utility, $\sum_{t=0}^{T-1} \beta^t u(c_t)$
3. **Behavior:** Choose $c_0, c_1, \dots, c_{T-1}$ *simultaneously*
4. **Solution:** Sequence of consumption *choices* $c_0^*, c_1^*, \dots, c_{T-1}^*$

- First order conditions:

$$\forall t : 0 = \beta^t u'(c_t) - \lambda R^{-t} \Leftrightarrow u'(c_t) = \lambda (\beta R)^{-t}$$

$$\mathcal{L} = \sum_{t=0}^{T-1} \beta^t u(c_t) + \lambda \left[s_0 + h_0 - \sum_{t=0}^{T-1} R^{-t} c_t \right]$$

- It is a *static* problem:

1. **Information:** s_0 and h_0 are known at $t = 0$
2. **Target:** Discounted utility, $\sum_{t=0}^{T-1} \beta^t u(c_t)$
3. **Behavior:** Choose $c_0, c_1, \dots, c_{T-1}$ *simultaneously*
4. **Solution:** Sequence of consumption *choices* $c_0^*, c_1^*, \dots, c_{T-1}^*$

- First order conditions:

$$\forall t : 0 = \beta^t u'(c_t) - \lambda R^{-t} \Leftrightarrow u'(c_t) = \lambda (\beta R)^{-t}$$

- Euler-equation for $k \in \{1, 2, \dots\}$:

$$\frac{u'(c_t)}{u'(c_{t+k})} = \frac{\lambda (\beta R)^{-t}}{\lambda (\beta R)^{-(t+k)}} = (\beta R)^k$$

- **CRRA:** $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$ imply Euler-equation

$$\frac{c_0^{-\sigma}}{c_t^{-\sigma}} = (\beta R)^t \Leftrightarrow c_t = (\beta R)^{\frac{t}{\sigma}} c_0$$

Constant consumption if:

1. $\beta R = 1$
2. $\frac{1}{\sigma} \rightarrow 0$ (zero intertemporal elasticity of substitution)

Consumption choice

- **CRRA:** $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$ imply Euler-equation

$$\frac{c_0^{-\sigma}}{c_t^{-\sigma}} = (\beta R)^t \Leftrightarrow c_t = (\beta R)^{\frac{t}{\sigma}} c_0$$

Constant consumption if:

1. $\beta R = 1$
2. $\frac{1}{\sigma} \rightarrow 0$ (zero intertemporal elasticity of substitution)

- Insert **Euler** into **IBC** to get consumption choice

$$\sum_{t=0}^{T-1} \left((\beta R)^{1/\sigma} R^{-1} \right)^t c_0 = (s_0 + h_0) \Leftrightarrow$$

$$c_0^* = \frac{1 - (\beta R)^{1/\sigma} R^{-1}}{1 - ((\beta R)^{1/\sigma} R^{-1})^T} (s_0 + h_0)$$

Infinite horizon: $T \rightarrow \infty$

- **Infinite horizon** for $(\beta R)^{1/\sigma} R^{-1} < 1$:

Let $T \rightarrow \infty$ to get

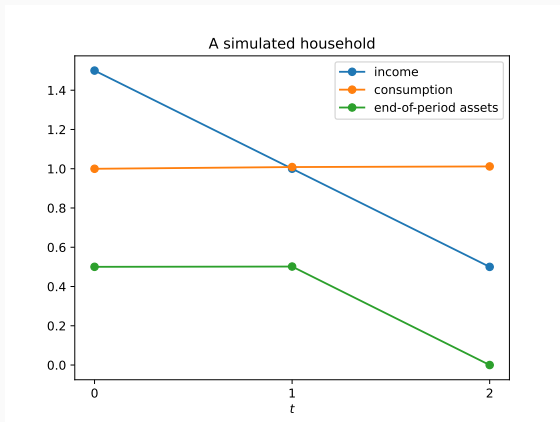
$$c_0^* = \left(1 - \frac{(\beta R)^{1/\sigma}}{R}\right) (s_0 + h_0)$$

Assuming $\beta R = 1$, this becomes

$$c_0 = \frac{r}{1+r} (s_0 + h_0)$$

$\Rightarrow$ Consumption is the annuity value of total wealth (perfect consumption smoothing!)

Consumption smoothing



Income falls by two thirds, consumption stays constant. The household saves while income is high and runs the savings down when it is low. ($T = 3$, $a_{-1} = 0$, from 02_vfi_pih.ipynb)

Propensities to consume ($\beta R = 1$, $z_t = 1$)

Assume $\beta R = 1$ and $z_t = 1$, then

$$c_0^* = ra_{-1} + w$$

MPC for different types of shocks (exact derivatives):

1. MPC of *windfall* income: $\frac{\partial c_0}{\partial s_0} = \frac{r}{1+r} \approx r\%$

Propensities to consume ($\beta R = 1$, $z_t = 1$)

Assume $\beta R = 1$ and $z_t = 1$, then

$$c_0^* = ra_{-1} + w$$

MPC for different types of shocks (exact derivatives):

1. MPC of *windfall* income: $\frac{\partial c_0}{\partial s_0} = \frac{r}{1+r} \approx r\%$
2. MPC of *future* income change: $\frac{\partial c_0}{\partial wz_t} = \frac{r}{1+r} (1+r)^{-t} \rightarrow$ even smaller

Propensities to consume ($\beta R = 1$, $z_t = 1$)

Assume $\beta R = 1$ and $z_t = 1$, then

$$c_0^* = ra_{-1} + w$$

MPC for different types of shocks (exact derivatives):

1. MPC of *windfall* income: $\frac{\partial c_0}{\partial s_0} = \frac{r}{1+r} \approx r\%$
2. MPC of *future* income change: $\frac{\partial c_0}{\partial wz_t} = \frac{r}{1+r} (1+r)^{-t} \rightarrow$ even smaller
3. MPC of *permanent* income change: $\frac{\partial c_0}{\partial w} = 1$

Connection to the representative agent

- The consumption function is *linear* in resources

$$c_0^* = \kappa (s_0 + h_0), \quad \kappa \equiv 1 - \frac{(\beta R)^{1/\sigma}}{R}$$

with the *same* κ for every household

Connection to the representative agent

- The consumption function is *linear* in resources

$$c_0^* = \kappa (s_0 + h_0), \quad \kappa \equiv 1 - \frac{(\beta R)^{1/\sigma}}{R}$$

with the *same* κ for every household

- **Exact aggregation:** sum over households i

$$\sum_i c_{0,i}^* = \kappa \left(\sum_i s_{0,i} + \sum_i h_{0,i} \right)$$

Connection to the representative agent

- The consumption function is *linear* in resources

$$c_0^* = \kappa (s_0 + h_0), \quad \kappa \equiv 1 - \frac{(\beta R)^{1/\sigma}}{R}$$

with the *same* κ for every household

- **Exact aggregation:** sum over households i

$$\sum_i c_{0,i}^* = \kappa \left(\sum_i s_{0,i} + \sum_i h_{0,i} \right)$$

- **Implication:** aggregate consumption depends on *aggregate* wealth and *aggregate* income only. The distribution does not enter, and a representative agent gets the aggregates exactly right.

Connection to the representative agent

- The consumption function is *linear* in resources

$$c_0^* = \kappa (s_0 + h_0), \quad \kappa \equiv 1 - \frac{(\beta R)^{1/\sigma}}{R}$$

with the *same* κ for every household

- **Exact aggregation:** sum over households i

$$\sum_i c_{0,i}^* = \kappa \left(\sum_i s_{0,i} + \sum_i h_{0,i} \right)$$

- **Implication:** aggregate consumption depends on *aggregate* wealth and *aggregate* income only. The distribution does not enter, and a representative agent gets the aggregates exactly right.
- **This is why the course exists:** *heterogeneity matters for aggregates only once this linearity breaks.*

Dynamic programming

Why we need a different method

- **What made the sequence approach work:**
 1. Substitute the budget constraint forward into a single IBC
 2. Impose one terminal condition
 3. Solve the FOCs for the whole path at once

Why we need a different method

- **What made the sequence approach work:**
 1. Substitute the budget constraint forward into a single IBC
 2. Impose one terminal condition
 3. Solve the FOCs for the whole path at once
- **Dynamic programming:** solve the problem *recursively*, one period at a time

Why we need a different method

- **What made the sequence approach work:**
 1. Substitute the budget constraint forward into a single IBC
 2. Impose one terminal condition
 3. Solve the FOCs for the whole path at once
- **Dynamic programming:** solve the problem *recursively*, one period at a time
- Will be very useful when we introduce risk and occasionally binding constraints

Dynamic solution: Bellman's Principle of Optimality

- **Go back to a finite horizon** with T periods, and solve it backwards from the terminal condition $v_T(\bullet) = 0$

Dynamic solution: Bellman's Principle of Optimality

- **Go back to a finite horizon** with T periods, and solve it backwards from the terminal condition $v_T(\bullet) = 0$
- **Principle:** *whatever the initial state and initial decision, the remaining decisions must be optimal given the state that results from the first decision*

Dynamic solution: Bellman's Principle of Optimality

- **Go back to a finite horizon** with T periods, and solve it backwards from the terminal condition $v_T(\bullet) = 0$
- **Principle:** *whatever the initial state and initial decision, the remaining decisions must be optimal given the state that results from the first decision*
- **Value function, v_t :** defined *recursively* from $v_T(\bullet) = 0$

$$v_t(a_{t-1}) = \max_{c_t} u(c_t) + \beta v_{t+1}(a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t, \quad a_{T-1} \geq 0$$

Dynamic solution: Bellman's Principle of Optimality

- **Go back to a finite horizon** with T periods, and solve it backwards from the terminal condition $v_T(\bullet) = 0$
- **Principle:** *whatever the initial state and initial decision, the remaining decisions must be optimal given the state that results from the first decision*
- **Value function, v_t :** defined *recursively* from $v_T(\bullet) = 0$

$$v_t(a_{t-1}) = \max_{c_t} u(c_t) + \beta v_{t+1}(a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t, \quad a_{T-1} \geq 0$$

- **Policy function, $c_t^*(a_{t-1})$:** we are looking for a **function** of **state variables**, instead of a **sequence**

Dynamic solution: Bellman's Principle of Optimality

- **Go back to a finite horizon** with T periods, and solve it backwards from the terminal condition $v_T(\bullet) = 0$
- **Principle:** *whatever the initial state and initial decision, the remaining decisions must be optimal given the state that results from the first decision*
- **Value function, v_t :** defined *recursively* from $v_T(\bullet) = 0$

$$v_t(a_{t-1}) = \max_{c_t} u(c_t) + \beta v_{t+1}(a_t)$$

$$\text{s.t. } a_t = (1+r)a_{t-1} + wz_t - c_t, \quad a_{T-1} \geq 0$$

- **Policy function, $c_t^*(a_{t-1})$:** we are looking for a **function** of **state variables**, instead of a **sequence**
- **Euler-equation** (interior choice):
 1. FOC: $c_t^{-\sigma} = \beta v_{t+1,a}(a_t)$
 2. Envelope: $v_{t,a}(a_{t-1}) = (1+r)c_t^{-\sigma}$ (fix c_t)
 3. Combine, shifting the envelope forward: $c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}$

$$v(a_{t-1}) = \max_{c_t} u(c_t) + \beta v(a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t, \quad a_t \geq 0$$

- **Contraction mapping result:** *If β is low enough (strong enough impatience) then the value and policy functions converge to $v(a_{t-1})$ and $c^*(a_{t-1})$ for large enough T*

$$v(a_{t-1}) = \max_{c_t} u(c_t) + \beta v(a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t, \quad a_t \geq 0$$

- **Contraction mapping result:** *If β is low enough (strong enough impatience) then the value and policy functions converge to $v(a_{t-1})$ and $c^*(a_{t-1})$ for large enough T*
- Problem becomes **stationary**

$$\begin{aligned} v(a_{t-1}) &= \max_{c_t} u(c_t) + \beta v(a_t) \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t, \quad a_t \geq 0 \end{aligned}$$

- **Contraction mapping result:** *If β is low enough (strong enough impatience) then the value and policy functions converge to $v(a_{t-1})$ and $c^*(a_{t-1})$ for large enough T*
- Problem becomes **stationary**
- **In practice:** for an infinite-horizon problem
 1. Make an arbitrary initial guess (e.g. $v_T = 0$)
 2. Iterate the backward step until the value (and/or policy) function does not change anymore, relative to a chosen tolerance

VFI



Three numerical problems

1. **Discretize the state:** a_{t-1} lives on a continuum, the computer needs a *grid*

Next lecture adds a fourth: *numerical integration*, once income is risky and the continuation value is an expectation.

Notebooks: 01_tools.ipynb, 02_vfi_pih.ipynb

Three numerical problems

1. **Discretize the state:** a_{t-1} lives on a continuum, the computer needs a *grid*
2. **Approximate off the grid:** v_{t+1} is known only at grid points, but $a_t = (1 + r)a_{t-1} + wz_t - c_t$ lands between them

Next lecture adds a fourth: *numerical integration*, once income is risky and the continuation value is an expectation.

Notebooks: 01_tools.ipynb, 02_vfi_pih.ipynb

Three numerical problems

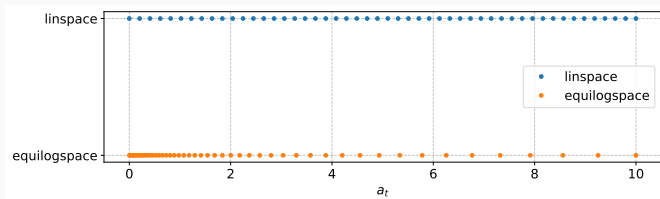
1. **Discretize the state:** a_{t-1} lives on a continuum, the computer needs a *grid*
2. **Approximate off the grid:** v_{t+1} is known only at grid points, but $a_t = (1 + r)a_{t-1} + wz_t - c_t$ lands between them
3. **Maximize:** find c_t^* at every grid point

Next lecture adds a fourth: *numerical integration*, once income is risky and the continuation value is an expectation.

Notebooks: 01_tools.ipynb, 02_vfi_pih.ipynb

- **Discretization:** the state variable belongs to a discrete set $\equiv$ a *grid*

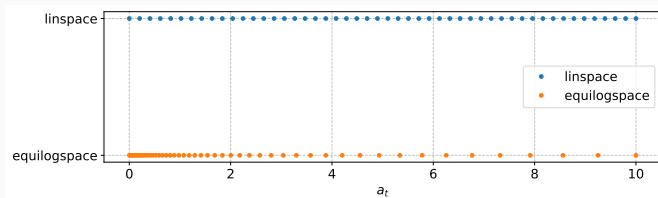
$$a_t \in \mathcal{G}_a = \{a^0, a^1, \dots, a^{\#_a-1}\}, \quad a^0 = \underline{a}$$



- **Discretization:** the state variable belongs to a discrete set $\equiv$ a *grid*

$$a_t \in \mathcal{G}_a = \{a^0, a^1, \dots, a^{\#_a-1}\}, \quad a^0 = \underline{a}$$

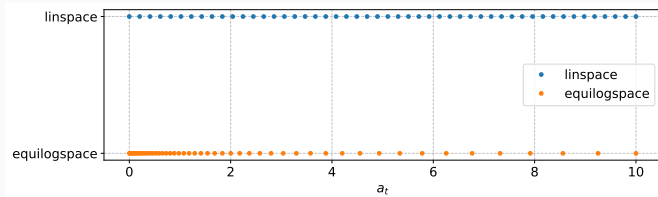
- **Not equally spaced:** the policy function bends most near the borrowing constraint, so that is where the points should be



- **Discretization:** the state variable belongs to a discrete set $\equiv$ a *grid*

$$a_t \in \mathcal{G}_a = \{a^0, a^1, \dots, a^{\#_a-1}\}, \quad a^0 = \underline{a}$$

- **Not equally spaced:** the policy function bends most near the borrowing constraint, so that is where the points should be
- **ConSav:** `grids.nonlinspace`, `grids.equilogspace`



- **Linear interpolation** (function approximation):

1. Assume v_{t+1} is known on $\mathcal{G}_a$
2. Evaluate $v_{t+1}(a)$ for arbitrary a by

$$\begin{aligned}\check{v}_{t+1}(a) &= \text{baseline} + \text{slope} \times \text{distance} \\ &= v_{t+1}(a^\iota) + \omega(a - a^\iota)\end{aligned}$$

where

$$\omega \equiv \frac{v_{t+1}(a^{\iota+1}) - v_{t+1}(a^\iota)}{a^{\iota+1} - a^\iota}$$

$$\iota \equiv \text{largest } i_a \in \{0, 1, \dots, \#_a - 2\} \text{ such that } a^{i_a} \leq a$$

- **Linear interpolation** (function approximation):

1. Assume v_{t+1} is known on $\mathcal{G}_a$
2. Evaluate $v_{t+1}(a)$ for arbitrary a by

$$\begin{aligned}\check{v}_{t+1}(a) &= \text{baseline} + \text{slope} \times \text{distance} \\ &= v_{t+1}(a^\iota) + \omega(a - a^\iota)\end{aligned}$$

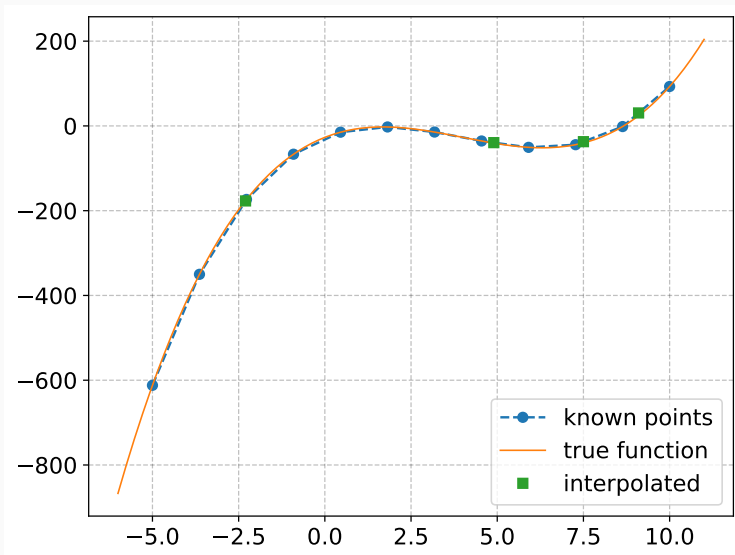
where

$$\omega \equiv \frac{v_{t+1}(a^{\iota+1}) - v_{t+1}(a^\iota)}{a^{\iota+1} - a^\iota}$$

$$\iota \equiv \text{largest } i_a \in \{0, 1, \dots, \#_a - 2\} \text{ such that } a^{i_a} \leq a$$

- **ConSav:** `linear_interp.interp_1d`

Linear interpolation



- **Grid search:** try $\#_c$ values of c_t , keep the best
 - Robust, but the accuracy is the spacing of the trial grid

- **Grid search:** try $\#_c$ values of c_t , keep the best
 - Robust, but the accuracy is the spacing of the trial grid
- **Golden section search:** keep a bracket $[\underline{c}, \overline{c}]$ known to contain the maximum, place two interior points, discard the part that cannot contain it
 - Bracket shrinks by $1/\varphi \approx 0.618$ per iteration
 - Requires the objective to be *unimodal*

- **Grid search:** try $\#_c$ values of c_t , keep the best
 - Robust, but the accuracy is the spacing of the trial grid
- **Golden section search:** keep a bracket $[\underline{c}, \overline{c}]$ known to contain the maximum, place two interior points, discard the part that cannot contain it
 - Bracket shrinks by $1/\varphi \approx 0.618$ per iteration
 - Requires the objective to be *unimodal*
- **Why derivative-free?** $\check{v}_{t+1}$ is piecewise linear, so its derivative is a step function

- **Grid search:** try $\#_c$ values of c_t , keep the best
 - Robust, but the accuracy is the spacing of the trial grid
- **Golden section search:** keep a bracket $[\underline{c}, \overline{c}]$ known to contain the maximum, place two interior points, discard the part that cannot contain it
 - Bracket shrinks by $1/\varphi \approx 0.618$ per iteration
 - Requires the objective to be *unimodal*
- **Why derivative-free?** $\check{v}_{t+1}$ is piecewise linear, so its derivative is a step function
- **ConSav:** `golden_section_search.optimizer`
Note: it *minimizes*, so pass the negative of the value

Value function iteration (VFI)

- Maximize the value-of-choice:

$$v_t(a_{i_{a-}}) = \max_{c_t} v_t(a_{i_{a-}} | c_t)$$

$$\text{with } c_t \in [0, (1+r)a_{i_{a-}} + wz_t - \underline{a}]$$

$$v_t(a^{i_{a-}} | c_t) = u(c_t) + \beta \check{v}_{t+1}(a_t)$$

$$\text{with } a_t = (1+r)a^{i_{a-}} + wz_t - c_t$$

Value function iteration (VFI)

- **Maximize the value-of-choice:**

$$v_t(a_{i_{a-}}) = \max_{c_t} v_t(a_{i_{a-}} | c_t)$$

with $c_t \in [0, (1+r)a_{i_{a-}} + wz_t - \underline{a}]$

$$v_t(a^{i_{a-}} | c_t) = u(c_t) + \beta \check{v}_{t+1}(a_t)$$

with $a_t = (1+r)a^{i_{a-}} + wz_t - c_t$

- **Inner loop:** for each grid point in $\mathcal{G}_a$ find $c_t^*(a_{t-1})$, and therefore $v_t(a_{t-1})$, with a *numerical optimizer*

Value function iteration (VFI)

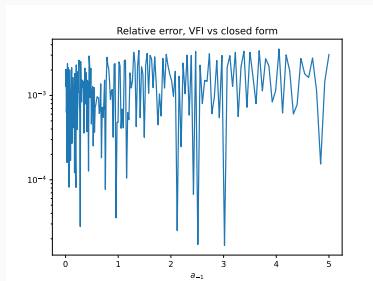
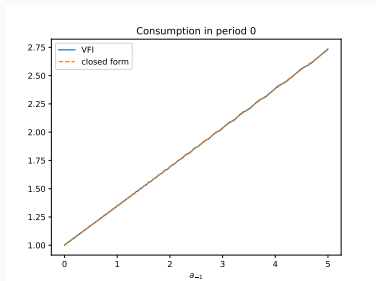
- **Maximize the value-of-choice:**

$$v_t(a_{i_{a-}}) = \max_{c_t} v_t(a_{i_{a-}} | c_t)$$
$$\text{with } c_t \in [0, (1+r)a_{i_{a-}} + wz_t - \underline{a}]$$

$$v_t(a^{i_{a-}} | c_t) = u(c_t) + \beta \check{v}_{t+1}(a_t)$$
$$\text{with } a_t = (1+r)a^{i_{a-}} + wz_t - c_t$$

- **Inner loop:** for each grid point in $\mathcal{G}_a$ find $c_t^*(a_{t-1})$, and therefore $v_t(a_{t-1})$, with a *numerical optimizer*
- **Outer loop:** backwards from $t = T - 1$ (note $v_T = 0$, or known)

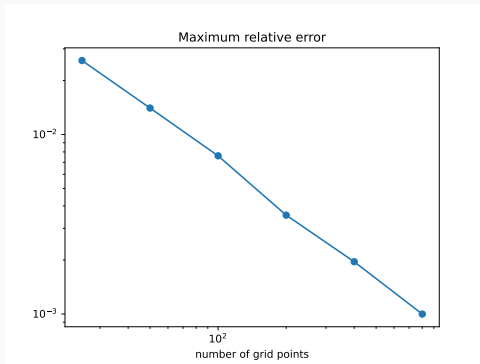
It works



The only model in the course with a known answer. $T = 3$,
 $\beta = 0.99$, $r = 0.02$, $\sigma = 2$, falling income.

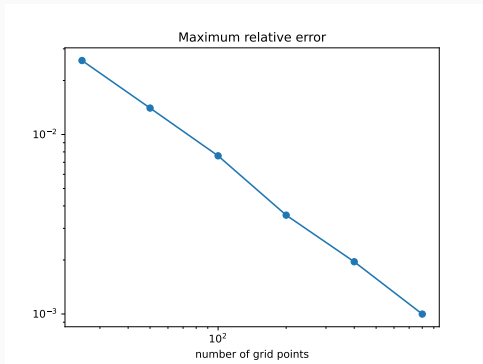
Maximum relative error: 3.6×10^{-3} with $\#_a = 200$.

Accuracy and the number of grid points



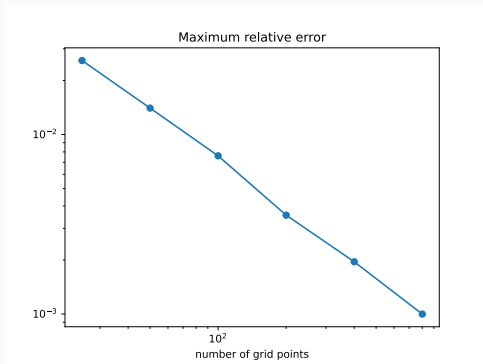
- Doubling the grid roughly halves the error

Accuracy and the number of grid points



- Doubling the grid roughly halves the error
- The bottleneck is interpolating v_{t+1} , which is strongly curved

Accuracy and the number of grid points



- Doubling the grid roughly halves the error
- The bottleneck is interpolating v_{t+1} , which is strongly curved
- $\Rightarrow$ *three more digits would need a grid a thousand times finer*

Buffer-stock

MPCs in the data are high

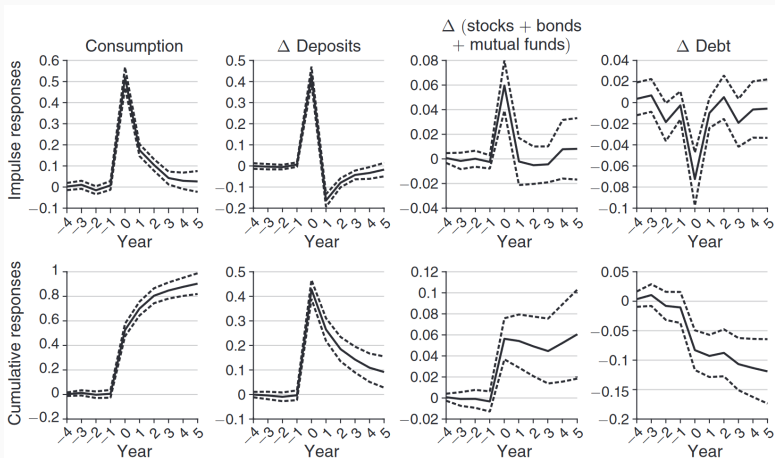


FIGURE 2. DYNAMIC HOUSEHOLD RESPONSES TO LOTTERY PRIZES

Source: Fagereng et al. (2021). **PIH** says $\frac{r}{1+r} \approx 2\%$.

What is missing from the model

- **Income risk** with a *prudent* household
⇒ precautionary saving, relatively larger for the asset poor

What is missing from the model

- **Income risk** with a *prudent* household
⇒ precautionary saving, relatively larger for the asset poor
- **A borrowing constraint that can bind at any date**
⇒ MPC of one for the constrained, and a kink in the consumption function

What is missing from the model

- **Income risk** with a *prudent* household
⇒ precautionary saving, relatively larger for the asset poor
- **A borrowing constraint that can bind at any date**
⇒ MPC of one for the constrained, and a kink in the consumption function
- **Both break the *linear* consumption function**
⇒ exact aggregation fails, and the distribution starts to matter

What is missing from the model

- **Income risk** with a *prudent* household
⇒ precautionary saving, relatively larger for the asset poor
- **A borrowing constraint that can bind at any date**
⇒ MPC of one for the constrained, and a kink in the consumption function
- **Both break the *linear* consumption function**
⇒ exact aggregation fails, and the distribution starts to matter
- **The price:** no closed form.
That is why we spent the last lecture on numerical methods.

Uncertainty and a borrowing constraint

$$v_0(z_0, a_{-1}) = \max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$z_t = \rho z_{t-1} + \sigma^z \epsilon_t$$

$$a_t \geq -wb$$

$$\lim_{t \rightarrow \infty} (1 + r)^{-t} a_t \geq 0 \quad [\text{No-Ponzi game}]$$

- **Stochastic income** from AR-1 process

Uncertainty and a borrowing constraint

$$v_0(z_0, a_{-1}) = \max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$z_t = \rho z_{t-1} + \sigma^z \epsilon_t$$

$$a_t \geq -wb$$

$$\lim_{t \rightarrow \infty} (1 + r)^{-t} a_t \geq 0 \quad [\text{No-Ponzi game}]$$

- **Stochastic income** from AR-1 process
- **Ad-hoc borrowing limit:** debt of at most $b \geq 0$, in units of the wage

Uncertainty and a borrowing constraint

$$v_0(z_0, a_{-1}) = \max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$z_t = \rho z_{t-1} + \sigma^z \epsilon_t$$

$$a_t \geq -wb$$

$$\lim_{t \rightarrow \infty} (1 + r)^{-t} a_t \geq 0 \quad [\text{No-Ponzi game}]$$

- **Stochastic income** from AR-1 process
- **Ad-hoc borrowing limit:** debt of at most $b \geq 0$, in units of the wage
- **A true dynamic problem:**
 1. **Information:** z_t is revealed period-by-period
 2. **Target:** Expected discounted utility, $\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$
 3. **Behavior:** Choose c_t *sequentially* as information is revealed
 4. **Solution:** Consumption *function*, $c^*(z_t, a_{t-1})$

$$\begin{aligned} v(z_t, a_{t-1}) &= \max_{c_t} u(c_t) + \beta \mathbb{E}_t [v(z_{t+1}, a_t)] \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a} \end{aligned}$$

1. **State variables:** z_t and a_{t-1}

Compare with last lecture: the *entire* cost of adding risk is one extra state variable and one expectation operator.

Note: no time subscript on v . The problem is stationary, so the policy function is the same in every period.

$$\begin{aligned} v(z_t, a_{t-1}) &= \max_{c_t} u(c_t) + \beta \mathbb{E}_t [v(z_{t+1}, a_t)] \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a} \end{aligned}$$

1. **State variables:** z_t and a_{t-1}
2. **Control (choice) variable:** c_t

Compare with last lecture: the *entire* cost of adding risk is one extra state variable and one expectation operator.

Note: no time subscript on v . The problem is stationary, so the policy function is the same in every period.

$$\begin{aligned} v(z_t, a_{t-1}) &= \max_{c_t} u(c_t) + \beta \mathbb{E}_t [v(z_{t+1}, a_t)] \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a} \end{aligned}$$

1. **State variables:** z_t and a_{t-1}
2. **Control (choice) variable:** c_t
3. **Continuation value:** $\beta \mathbb{E}_t [v(z_{t+1}, a_t)]$

Compare with last lecture: the *entire* cost of adding risk is one extra state variable and one expectation operator.

Note: no time subscript on v . The problem is stationary, so the policy function is the same in every period.

$$v(z_t, a_{t-1}) = \max_{c_t} u(c_t) + \beta \mathbb{E}_t [v(z_{t+1}, a_t)]$$
$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a}$$

1. **State variables:** z_t and a_{t-1}
2. **Control (choice) variable:** c_t
3. **Continuation value:** $\beta \mathbb{E}_t [v(z_{t+1}, a_t)]$
4. **Parameters:** r , w , and stuff in $u(\bullet)$

Compare with last lecture: the *entire* cost of adding risk is one extra state variable and one expectation operator.

Note: no time subscript on v . The problem is stationary, so the policy function is the same in every period.

- **Substitution** still implies (path-by-path, for any realization of z_t):

$$R^{-(T-1)}a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t}c_t = s_0 + h_0$$

- **Substitution** still implies (path-by-path, for any realization of z_t):

$$R^{-(T-1)}a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t}c_t = s_0 + h_0$$

- **What if $T \rightarrow \infty$?** We must have $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} = 0$
 1. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} > 0$: Consumption can be increased
 2. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} < 0$: Violates No-Ponzi game condition

- **Substitution** still implies (path-by-path, for any realization of z_t):

$$R^{-(T-1)}a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t}c_t = s_0 + h_0$$

- **What if $T \rightarrow \infty$?** We must have $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} = 0$
 1. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} > 0$: Consumption can be increased
 2. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} < 0$: Violates No-Ponzi game condition
- For $T \rightarrow \infty$ we have the **IBC**:

$$\sum_{t=0}^{\infty} R^{-t}c_t = Ra_{-1} + \sum_{t=0}^{\infty} R^{-t}wz_t$$

Natural borrowing limit

- Denote **minimum possible productivity** by $\underline{z}$

Natural borrowing limit

- Denote **minimum possible productivity** by $\underline{z}$
- **Consumption must be non-negative** $\Rightarrow$
interest payments must be less than minimum income

$$c_t \geq 0 \Rightarrow r(-a_t) \leq w\underline{z} \Leftrightarrow a_t \geq -\frac{w\underline{z}}{r}$$

If debt was larger it would in the worst case ($\forall z_t = \underline{z}$) grow without bound even with zero consumption ($\forall c_t = 0$)

$$a_0 = -\frac{w\underline{z}}{r} - \Delta$$

$$a_1 = (1+r)a_0 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)\Delta$$

$$a_2 = (1+r)a_1 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)^2\Delta$$

$$\vdots$$

Natural borrowing limit

- Denote **minimum possible productivity** by $\underline{z}$
- Consumption must be non-negative** $\Rightarrow$
interest payments must be less than minimum income

$$c_t \geq 0 \Rightarrow r(-a_t) \leq w\underline{z} \Leftrightarrow a_t \geq -\frac{w\underline{z}}{r}$$

If debt was larger it would in the worst case ($\forall z_t = \underline{z}$) grow without bound even with zero consumption ($\forall c_t = 0$)

$$a_0 = -\frac{w\underline{z}}{r} - \Delta$$

$$a_1 = (1+r)a_0 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)\Delta$$

$$a_2 = (1+r)a_1 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)^2\Delta$$

$\vdots$

- Natural borrowing constraint:** $a_t \geq \underline{a} = -w \min \left\{ b, \frac{\underline{z}}{r} \right\}$

Euler-equation with an occasionally binding constraint

- **Attach a multiplier $\mu_t \geq 0$ to $a_t \geq \underline{a}$**

$$\text{FOC: } c_t^{-\sigma} = \beta \mathbb{E}_t [v_a(z_{t+1}, a_t)] + \mu_t$$

$$\text{Slackness: } \mu_t (a_t - \underline{a}) = 0$$

$$\text{Envelope: } v_a(z_t, a_{t-1}) = (1 + r)c_t^{-\sigma}$$

Euler-equation with an occasionally binding constraint

- **Attach a multiplier $\mu_t \geq 0$ to $a_t \geq \underline{a}$**

$$\text{FOC: } c_t^{-\sigma} = \beta \mathbb{E}_t [v_a(z_{t+1}, a_t)] + \mu_t$$

$$\text{Slackness: } \mu_t (a_t - \underline{a}) = 0$$

$$\text{Envelope: } v_a(z_t, a_{t-1}) = (1 + r)c_t^{-\sigma}$$

- **Two cases:**

1. **Unconstrained** ($a_t > \underline{a}$, so $\mu_t = 0$):

$$c_t^{-\sigma} = \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}]$$

2. **Constrained** ($a_t = \underline{a}$, so $\mu_t > 0$):

$$c_t^{-\sigma} > \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}] \quad \text{and} \quad c_t = (1 + r)a_{t-1} + wz_t - \underline{a}$$

Euler-equation with an occasionally binding constraint

- Attach a multiplier $\mu_t \geq 0$ to $a_t \geq \underline{a}$

$$\text{FOC: } c_t^{-\sigma} = \beta \mathbb{E}_t [v_a(z_{t+1}, a_t)] + \mu_t$$

$$\text{Slackness: } \mu_t (a_t - \underline{a}) = 0$$

$$\text{Envelope: } v_a(z_t, a_{t-1}) = (1 + r)c_t^{-\sigma}$$

- Two cases:

1. Unconstrained ($a_t > \underline{a}$, so $\mu_t = 0$):

$$c_t^{-\sigma} = \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}]$$

2. Constrained ($a_t = \underline{a}$, so $\mu_t > 0$):

$$c_t^{-\sigma} > \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}] \quad \text{and} \quad c_t = (1 + r)a_{t-1} + wz_t - \underline{a}$$

- Why an inequality? Lowering c_t is always feasible, raising it is not. The household would like to borrow and cannot.

- **Prudence:** $u'''(c) > 0$, i.e. marginal utility is *convex*
CRRA has $u'''(c) = \sigma(\sigma + 1)c^{-\sigma-2} > 0$

Prudence and precautionary saving

- **Prudence:** $u'''(c) > 0$, i.e. marginal utility is *convex*
CRRA has $u'''(c) = \sigma(\sigma + 1)c^{-\sigma-2} > 0$
- **Jensen on the Euler-equation** (take $\beta R = 1$, unconstrained):

$$u'(c_t) = \mathbb{E}_t [u'(c_{t+1})] > u'(\mathbb{E}_t [c_{t+1}]) \Leftrightarrow \mathbb{E}_t [c_{t+1}] > c_t$$

Prudence and precautionary saving

- **Prudence:** $u'''(c) > 0$, i.e. marginal utility is *convex*
CRRA has $u'''(c) = \sigma(\sigma + 1)c^{-\sigma-2} > 0$
- **Jensen on the Euler-equation** (take $\beta R = 1$, unconstrained):

$$u'(c_t) = \mathbb{E}_t [u'(c_{t+1})] > u'(\mathbb{E}_t [c_{t+1}]) \Leftrightarrow \mathbb{E}_t [c_{t+1}] > c_t$$

- **Interpretation:** the household *tilts consumption into the future* to build a buffer. Risk lowers consumption today even with no constraint.

Prudence and precautionary saving

- **Prudence:** $u'''(c) > 0$, i.e. marginal utility is *convex*
CRRA has $u'''(c) = \sigma(\sigma + 1)c^{-\sigma-2} > 0$
- **Jensen on the Euler-equation** (take $\beta R = 1$, unconstrained):

$$u'(c_t) = \mathbb{E}_t [u'(c_{t+1})] > u'(\mathbb{E}_t [c_{t+1}]) \Leftrightarrow \mathbb{E}_t [c_{t+1}] > c_t$$

- **Interpretation:** the household *tilts consumption into the future* to build a buffer. Risk lowers consumption today even with no constraint.
- **Two distinct channels, same direction:**
 1. **Risk $\times$ prudence:** precautionary saving
 2. **Borrowing constraint:** cannot smooth downwards

Special case I: Quadratic utility

- **Quadratic utility:** $u(c_t) = -\frac{1}{2}(\bar{c} - c)^2$ with $\beta R = 1$ and »large« $\bar{c}$

Special case I: Quadratic utility

- **Quadratic utility:** $u(c_t) = -\frac{1}{2}(\bar{c} - c)^2$ with $\beta R = 1$ and »large« $\bar{c}$
- **Euler-equation:** *Consumption = expected future consumption*

$$(\bar{c} - c_t) = \mathbb{E}_t[(\bar{c} - c_{t+k})] \Leftrightarrow c_t = \mathbb{E}_t[c_{t+k}]$$

Special case I: Quadratic utility

- **Quadratic utility:** $u(c_t) = -\frac{1}{2}(\bar{c} - c)^2$ with $\beta R = 1$ and »large« $\bar{c}$
- **Euler-equation:** *Consumption = expected future consumption*

$$(\bar{c} - c_t) = \mathbb{E}_t[(\bar{c} - c_{t+k})] \Leftrightarrow c_t = \mathbb{E}_t[c_{t+k}]$$

- Use **IBC** in expectation to get **consumption function**:

$$\sum_{s=0}^{\infty} R^{-s} \mathbb{E}_0[c_s] = Ra_{-1} + \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s] \Rightarrow$$
$$c^*(z_0, a_{-1}) = c_0 = ra_{-1} + \frac{r}{R} \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s]$$

where we formally disregard the borrowing constraint

Special case I: Quadratic utility

- **Quadratic utility:** $u(c_t) = -\frac{1}{2}(\bar{c} - c)^2$ with $\beta R = 1$ and »large« $\bar{c}$
- **Euler-equation:** *Consumption = expected future consumption*

$$(\bar{c} - c_t) = \mathbb{E}_t[(\bar{c} - c_{t+k})] \Leftrightarrow c_t = \mathbb{E}_t[c_{t+k}]$$

- Use **IBC** in expectation to get **consumption function**:

$$\sum_{s=0}^{\infty} R^{-s} \mathbb{E}_0[c_s] = Ra_{-1} + \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s] \Rightarrow$$
$$c^*(z_0, a_{-1}) = c_0 = ra_{-1} + \frac{r}{R} \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s]$$

where we formally disregard the borrowing constraint

- **Certainty equivalence:** *Only expected income matters.* No prudence: $u''' = 0$.

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}$

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}$
- **Productivity is absolute random walk:**

$$z_t = z_{t-1} + \psi_t$$

$$\psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}$
- **Productivity is absolute random walk:**

$$z_t = z_{t-1} + \psi_t$$

$$\psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

- **Consumption function (see proof):**

$$c^*(a_{t-1}, z_t) = ra_{t-1} + wz_t - \frac{\frac{1}{\alpha} \log(\beta R) + \frac{\alpha w^2 \sigma_\psi^2}{2}}{r}$$

where we formally disregard the borrowing constraint

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}$
- **Productivity is absolute random walk:**

$$z_t = z_{t-1} + \psi_t$$

$$\psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

- **Consumption function (see proof):**

$$c^*(a_{t-1}, z_t) = ra_{t-1} + wz_t - \frac{\frac{1}{\alpha} \log(\beta R) + \frac{\alpha w^2 \sigma_\psi^2}{2}}{r}$$

where we formally disregard the borrowing constraint

- **Precautionary saving:** $\sigma_\psi^2 \uparrow$ implies $c_t^* \downarrow$ for given z_t and a_{t-1}
 $\Rightarrow$ *accumulation of buffer-stock*

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c}$
- **Productivity is absolute random walk:**

$$z_t = z_{t-1} + \psi_t$$

$$\psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

- **Consumption function (see proof):**

$$c^*(a_{t-1}, z_t) = ra_{t-1} + wz_t - \frac{\frac{1}{\alpha} \log(\beta R) + \frac{\alpha w^2 \sigma_\psi^2}{2}}{r}$$

where we formally disregard the borrowing constraint

- **Precautionary saving:** $\sigma_\psi^2 \uparrow$ implies $c_t^* \downarrow$ for given z_t and a_{t-1}
 $\Rightarrow$ *accumulation of buffer-stock*
- **But:** the MPC is still *constant*. Heterogeneous MPCs need CRRA + a constraint, and then a computer.

- **Two forces on assets:**
 1. **Impatience** ($\beta R < 1$): pushes assets *down*
 2. **Prudence** $\times$ **risk**: pushes assets *up*

- **Two forces on assets:**
 1. **Impatience** ($\beta R < 1$): pushes assets *down*
 2. **Prudence** $\times$ **risk**: pushes assets *up*
- **Buffer-stock target:** they balance at a level a^* where $\mathbb{E}_t[a_t] = a_{t-1}$
 - Below a^* : the household builds its buffer
 - Above a^* : the household runs it down

- **Two forces on assets:**
 1. **Impatience** ($\beta R < 1$): pushes assets *down*
 2. **Prudence** $\times$ **risk**: pushes assets *up*
- **Buffer-stock target:** they balance at a level a^* where $\mathbb{E}_t[a_t] = a_{t-1}$
 - Below a^* : the household builds its buffer
 - Above a^* : the household runs it down
- **Carroll's impatience condition:** a target exists if the household is impatient enough to decumulate without risk, but prudent enough to save against it

- **Two forces on assets:**
 1. **Impatience** ($\beta R < 1$): pushes assets *down*
 2. **Prudence** $\times$ **risk**: pushes assets *up*
- **Buffer-stock target:** they balance at a level a^* where $\mathbb{E}_t[a_t] = a_{t-1}$
 - Below a^* : the household builds its buffer
 - Above a^* : the household runs it down
- **Carroll's impatience condition:** a target exists if the household is impatient enough to decumulate without risk, but prudent enough to save against it
- **What to look for in the numerical solution:**
 1. Consumption *concave* in assets
 2. MPC *decreasing* in assets, and one at the constraint
 3. Net savings crossing zero

- **Beginning-of-period value function**

$$\underline{v}(z_t, a_t) = \mathbb{E}[v(z_{t+1}, a_t) \mid z_t]$$

- **Beginning-of-period value function**

$$\underline{v}(z_t, a_t) = \mathbb{E}[v(z_{t+1}, a_t) | z_t]$$

- **Value function:**

$$v(z_t, a_{t-1}) = \max_{c_t} u(c_t) + \beta \underline{v}(z_t, a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a}$$

Recursive formulation with $\underline{v}$

- Beginning-of-period value function

$$\underline{v}(z_t, a_t) = \mathbb{E}[v(z_{t+1}, a_t) | z_t]$$

- Value function:

$$v(z_t, a_{t-1}) = \max_{c_t} u(c_t) + \beta \underline{v}(z_t, a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a}$$

- **Why do this?** *Only one interpolation per guess of c_t*
Integrating *inside* the maximization would cost $\#_z$ interpolations per guess.

Recursive formulation with $\underline{v}$

- **Beginning-of-period value function**

$$\underline{v}(z_t, a_t) = \mathbb{E}[v(z_{t+1}, a_t) | z_t]$$

- **Value function:**

$$v(z_t, a_{t-1}) = \max_{c_t} u(c_t) + \beta \underline{v}(z_t, a_t)$$

$$\text{s.t. } a_t = (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a}$$

- **Why do this?** *Only one interpolation per guess of c_t*
Integrating *inside* the maximization would cost $\#_z$ interpolations per guess.
- **This is the convention** GEModelTools **uses from the next lecture on**

What risk costs us numerically

- The state is now two-dimensional: $\mathcal{G}_z \times \mathcal{G}_a$ (tensor product)

What risk costs us numerically

- **The state is now two-dimensional:** $\mathcal{G}_z \times \mathcal{G}_a$ (tensor product)
- **The continuation value is an expectation.** Two new jobs:
 1. Turn the AR(1) into a *finite Markov chain*
 2. Turn $\mathbb{E}_t$ into a *sum*

$$\underline{v}(z^{i_z}, a_t) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} v(z^{i_{z+}}, a_t)$$

which is one matrix multiplication

What risk costs us numerically

- **The state is now two-dimensional:** $\mathcal{G}_z \times \mathcal{G}_a$ (tensor product)
- **The continuation value is an expectation.** Two new jobs:
 1. Turn the AR(1) into a *finite Markov chain*
 2. Turn $\mathbb{E}_t$ into a *sum*

$$\underline{v}(z^{i_z}, a_t) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} v(z^{i_{z+}}, a_t)$$

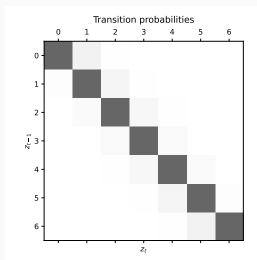
which is one matrix multiplication

- **Everything else from last lecture carries over unchanged:** grids, interpolation, the optimizer

Discretizing an AR(1)

- Income process:

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2), \quad \mathbb{E}[z_t] = 1$$

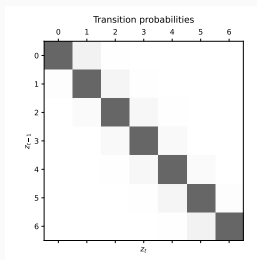


Discretizing an AR(1)

- Income process:

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2), \quad \mathbb{E}[z_t] = 1$$

- Tauchen (1986):** evenly spaced grid spanning the unconditional distribution, match the transition probabilities by integrating the normal density

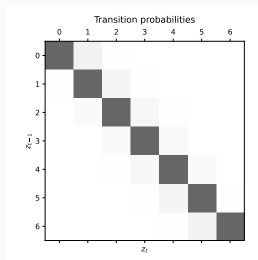


Discretizing an AR(1)

- Income process:

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2), \quad \mathbb{E}[z_t] = 1$$

- Tauchen (1986):** evenly spaced grid spanning the unconditional distribution, match the transition probabilities by integrating the normal density
- Rouwenhorst (1995):** build the transition matrix by recursion, matching the persistence and the unconditional variance *exactly*
 - Preferred at high ρ_z , where Tauchen needs many nodes

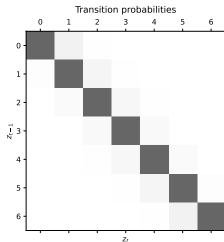


Discretizing an AR(1)

- Income process:

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2), \quad \mathbb{E}[z_t] = 1$$

- Tauchen (1986)**: evenly spaced grid spanning the unconditional distribution, match the transition probabilities by integrating the normal density
- Rouwenhorst (1995)**: build the transition matrix by recursion, matching the persistence and the unconditional variance *exactly*
 - Preferred at high ρ_z , where Tauchen needs many nodes
- ConSav**: `markov.log_rouwenhorst` `markov.tauchen`



- Maximize the value-of-choice:

$$v(z^{i_z}, a^{i_{a-}}) = \max_{c_t} v(z^{i_z}, a^{i_{a-}} | c_t)$$

$$\text{with } c_t \in [0, (1+r)a^{i_{a-}} + wz^{i_z} - \underline{a}]$$

$$v(z^{i_z}, a^{i_{a-}} | c_t) = u(c_t) + \beta \check{v}(z^{i_z}, a_t)$$

$$\text{with } a_t = (1+r)a^{i_{a-}} + wz^{i_z} - c_t$$

- Maximize the value-of-choice:

$$v(z^{i_z}, a^{i_{a-}}) = \max_{c_t} v(z^{i_z}, a^{i_{a-}} | c_t)$$
$$\text{with } c_t \in [0, (1+r)a^{i_{a-}} + wz^{i_z} - \underline{a}]$$

$$v(z^{i_z}, a^{i_{a-}} | c_t) = u(c_t) + \beta \underline{v}(z^{i_z}, a_t)$$
$$\text{with } a_t = (1+r)a^{i_{a-}} + wz^{i_z} - c_t$$

- Expectation step: $\underline{v} = \Pi_z v$, before the maximization

- **Maximize the value-of-choice:**

$$v(z^{i_z}, a^{i_{a-}}) = \max_{c_t} v(z^{i_z}, a^{i_{a-}} | c_t)$$
$$\text{with } c_t \in [0, (1+r)a^{i_{a-}} + wz^{i_z} - \underline{a}]$$

$$v(z^{i_z}, a^{i_{a-}} | c_t) = u(c_t) + \beta \underline{v}(z^{i_z}, a_t)$$
$$\text{with } a_t = (1+r)a^{i_{a-}} + wz^{i_z} - c_t$$

- **Expectation step:** $\underline{v} = \Pi_z v$, before the maximization
- **Inner loop:** over $\mathcal{G}_z \times \mathcal{G}_a$ instead of $\mathcal{G}_a$

- **Maximize the value-of-choice:**

$$v(z^{i_z}, a^{i_a-}) = \max_{c_t} v(z^{i_z}, a^{i_a-} | c_t)$$
$$\text{with } c_t \in [0, (1+r)a^{i_a-} + wz^{i_z} - \underline{a}]$$

$$v(z^{i_z}, a^{i_a-} | c_t) = u(c_t) + \beta \check{v}(z^{i_z}, a_t)$$
$$\text{with } a_t = (1+r)a^{i_a-} + wz^{i_z} - c_t$$

- **Expectation step:** $\underline{v} = \Pi_z v$, before the maximization
- **Inner loop:** over $\mathcal{G}_z \times \mathcal{G}_a$ instead of $\mathcal{G}_a$
- **Outer loop:** iterate until v stops moving, instead of backwards from T

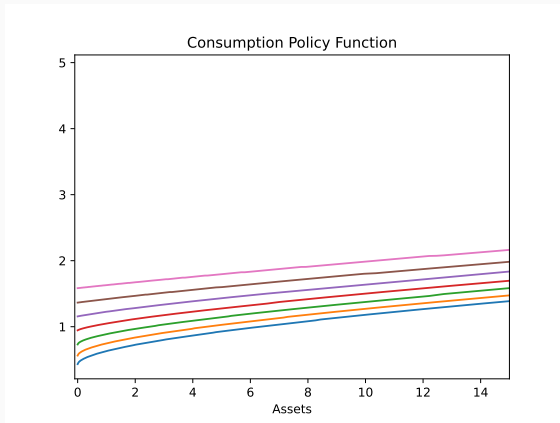
- **Maximize the value-of-choice:**

$$v(z^{i_z}, a^{i_a-}) = \max_{c_t} v(z^{i_z}, a^{i_a-} | c_t)$$
$$\text{with } c_t \in [0, (1+r)a^{i_a-} + wz^{i_z} - \underline{a}]$$

$$v(z^{i_z}, a^{i_a-} | c_t) = u(c_t) + \beta \check{v}(z^{i_z}, a_t)$$
$$\text{with } a_t = (1+r)a^{i_a-} + wz^{i_z} - c_t$$

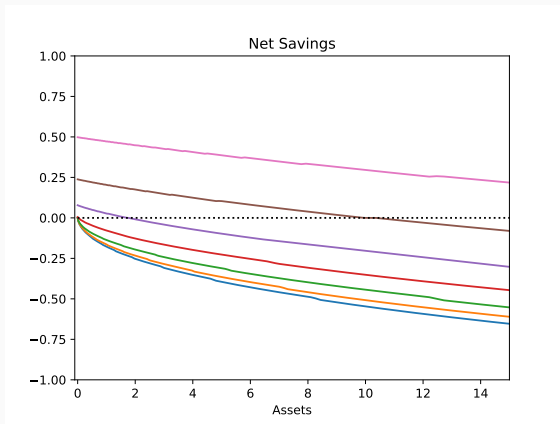
- **Expectation step:** $\underline{v} = \Pi_z v$, before the maximization
- **Inner loop:** over $\mathcal{G}_z \times \mathcal{G}_a$ instead of $\mathcal{G}_a$
- **Outer loop:** iterate until v stops moving, instead of backwards from T
- **Notebook:** 03_vfi_bufferstock.ipynb

Consumption function



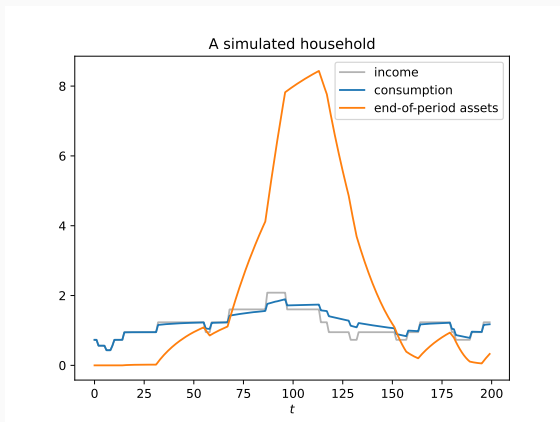
Concave in assets, and **kinked** where the borrowing constraint starts to bind. One line per productivity state.

Net savings and the buffer-stock target

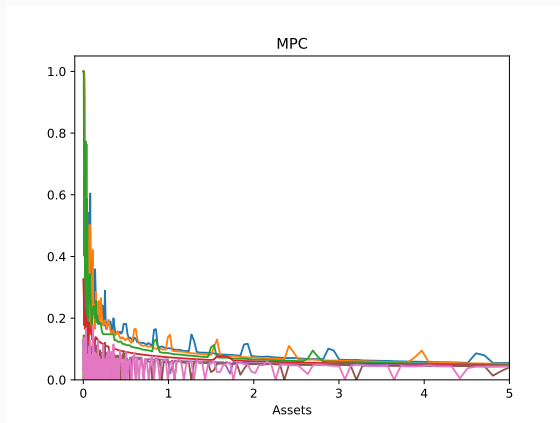


Decreasing in wealth and crossing zero: the crossing point is the target. It exists despite $\beta R < 1$.

One household, simulated

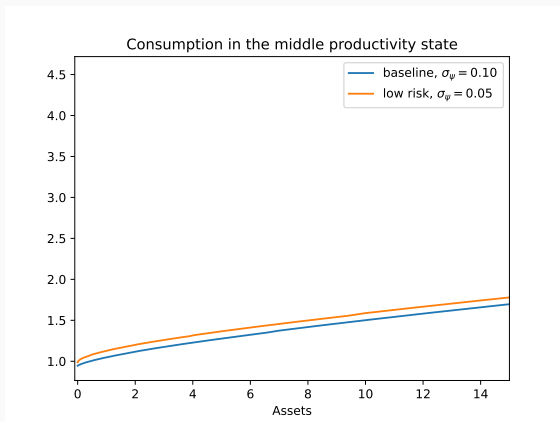


Consumption is smoother than income, but not flat. The buffer is built in good times and run down in bad ones; when it runs out, consumption tracks income one for one. (from `03_vfi_bufferstock.ipynb`)



One at the constraint, falling steeply with assets. Under the PIH it was $\frac{r}{1+r}$ for everybody.

How much of this is risk?



Halve σ_ψ and consumption rises everywhere. The gap is largest for the asset poor: everybody faces the same risk, but not everybody has a buffer against it.

- **Consumption lower than under PIH and concave in assets**

Intuition: *Precautionary saving motive is relatively larger for asset poor households because income risk is the same for everybody*

Implications:

1. Windfall gives safety and increases average consumption
⇒ MPC decreasing in assets
2. Attraction towards a buffer-stock target $a_t = a_{t-1}$ despite $\beta R < 1$
3. Larger effective discounting of future income
(extreme: no effect of future income changes if constrained before)

- **Value Function Iteration (VFI)**

1. Solves all consumption-saving models
2. Accurate with dense enough grids
3. Relatively simple code and easy to run in parallel
4. Finding optimal choices is the computational bottleneck (especially with multi-starts in non-convex models)
5. Interpolating a strongly curved v limits the accuracy

EGM



- **Replace numerical optimization with root-finding**

- **Replace numerical optimization with root-finding**
- **Time iteration:** For each a_{t-1} and z_t find c_t to solve the Euler-equation

$$c_t^{-\sigma} = \beta(1+r)\mathbb{E}_t[c_{t+1}^{-\sigma}]$$

Note: *Necessary and sufficient* (for interior choices, else $a_t = \underline{a}$)

- **Replace numerical optimization with root-finding**
- **Time iteration:** For each a_{t-1} and z_t find c_t to solve the Euler-equation

$$c_t^{-\sigma} = \beta(1+r)\mathbb{E}_t[c_{t+1}^{-\sigma}]$$

Note: *Necessary and sufficient* (for interior choices, else $a_t = \underline{a}$)

- **EGM:** No need for any numerical optimization *or* root-finding

Endogenous grid-point method (EGM)

1. Calculate **post-decision marginal value of cash**:

$$q(z^{i_z}, a^{i_a}) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} c_+^*(z^{i_{z+}}, a^{i_a})^{-\sigma}$$

Endogenous grid-point method (EGM)

1. Calculate **post-decision marginal value of cash**:

$$q(z^{i_z}, a^{i_a}) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} c_+^*(z^{i_{z+}}, a^{i_a})^{-\sigma}$$

2. **Invert Euler-equation**:

$$c(z^{i_z}, a^{i_a}) = (\beta(1+r)q(z^{i_z}, a^{i_a}))^{-\frac{1}{\sigma}}$$

Endogenous grid-point method (EGM)

1. Calculate **post-decision marginal value of cash**:

$$q(z^{i_z}, a^{i_a}) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} c_+^*(z^{i_{z+}}, a^{i_a})^{-\sigma}$$

2. **Invert Euler-equation**:

$$c(z^{i_z}, a^{i_a}) = (\beta(1+r)q(z^{i_z}, a^{i_a}))^{-\frac{1}{\sigma}}$$

3. **Endogenous cash-on-hand**:

$$m(z^{i_z}, a^{i_a}) = a^{i_a} + c(z^{i_z}, a^{i_a})$$

Endogenous grid-point method (EGM)

1. Calculate **post-decision marginal value of cash**:

$$q(z^{i_z}, a^{i_a}) = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} c_+^*(z^{i_{z+}}, a^{i_a})^{-\sigma}$$

2. **Invert Euler-equation**:

$$c(z^{i_z}, a^{i_a}) = (\beta(1+r)q(z^{i_z}, a^{i_a}))^{-\frac{1}{\sigma}}$$

3. **Endogenous cash-on-hand**:

$$m(z^{i_z}, a^{i_a}) = a^{i_a} + c(z^{i_z}, a^{i_a})$$

4. **Consumption function**: Calculate $m = (1+r)a^{i_a-} + wz^{i_z}$

- 4.1 Binding constraint: If $m \leq m(z^{i_z}, a^0)$ then

$$c^*(z^{i_z}, a^{i_a-}) = m - \underline{a}$$

- 4.2 Interior choice: Else

$$c^*(z^{i_z}, a^{i_a-}) = \text{interpolate } m(z^{i_z}, \bullet) \rightarrow c(z^{i_z}, \bullet) \text{ at } m$$

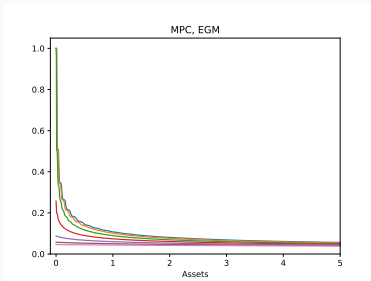
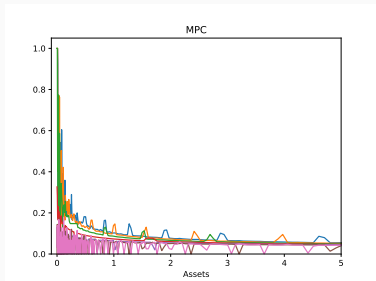
- **Fix a_t instead of m_t .** Then the expectation is taken *before* the choice, and the FOC inverts in closed form.
 1. No optimizer, no root-finder
 2. One interpolation per grid point

- **Fix a_t instead of m_t .** Then the expectation is taken *before* the choice, and the FOC inverts in closed form.
 1. No optimizer, no root-finder
 2. One interpolation per grid point
- **It also interpolates the *policy* function rather than the *value* function**
 - ⇒ much more accurate, because c^* is close to linear while v is strongly curved

- **Fix a_t instead of m_t .** Then the expectation is taken *before* the choice, and the FOC inverts in closed form.
 1. No optimizer, no root-finder
 2. One interpolation per grid point
- **It also interpolates the *policy* function rather than the *value* function**
⇒ much more accurate, because c^* is close to linear while v is strongly curved
- **Where it breaks:** non-concave problems and discrete choices, where the FOC is not sufficient
 - Fix: the *upper envelope* method, see lecture 13

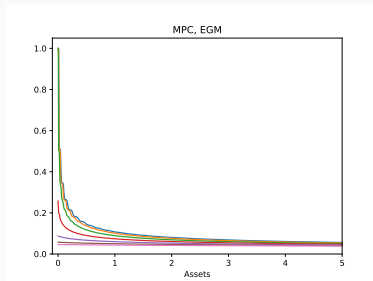
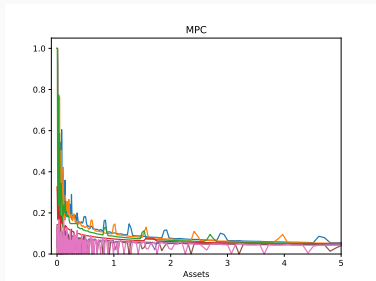
- **Fix a_t instead of m_t .** Then the expectation is taken *before* the choice, and the FOC inverts in closed form.
 1. No optimizer, no root-finder
 2. One interpolation per grid point
- **It also interpolates the *policy* function rather than the *value* function**
⇒ much more accurate, because c^* is close to linear while v is strongly curved
- **Where it breaks:** non-concave problems and discrete choices, where the FOC is not sufficient
 - Fix: the *upper envelope* method, see lecture 13
- **Notebook:** 04_egm.ipynb

VFI vs EGM



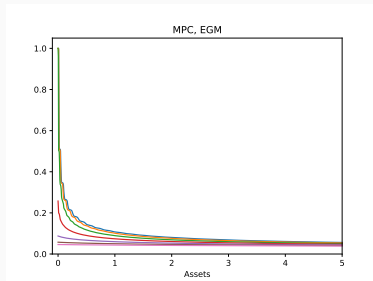
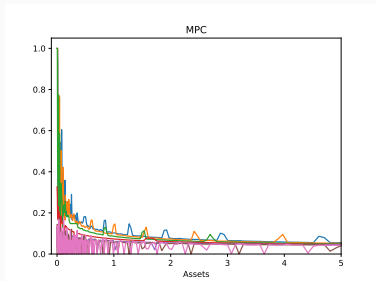
- Same model, same MPC, computed the same way (first difference of c along the grid)

VFI vs EGM



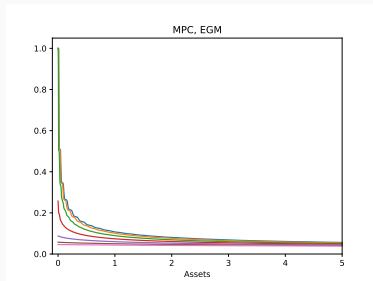
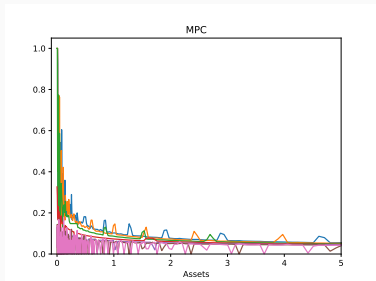
- Same model, same MPC, computed the same way (first difference of c along the grid)
- **Left, VFI:** ragged. $\check{v}$ is piecewise linear, so a_t gets pinned at grid points

VFI vs EGM



- Same model, same MPC, computed the same way (first difference of c along the grid)
- **Left, VFI:** ragged. $\check{v}$ is piecewise linear, so a_t gets pinned at grid points
- **Right, EGM:** clean

VFI vs EGM



- Same model, same MPC, computed the same way (first difference of c along the grid)
- **Left, VFI:** ragged. $\check{v}$ is piecewise linear, so a_t gets pinned at grid points
- **Right, EGM:** clean
- **On the deterministic problem:** EGM is exact at $\#_a = 25$, VFI needs hundreds of points for three digits

Simulation and aggregation

From policy functions to aggregates

- **What we have:** $c^*(z_t, a_{t-1})$ and $a^*(z_t, a_{t-1})$, what *one* household does in every state

From policy functions to aggregates

- **What we have:** $c^*(z_t, a_{t-1})$ and $a^*(z_t, a_{t-1})$, what *one* household does in every state
- **What the next lecture needs:** aggregates, e.g.

$$A_t^{hh} = \sum_{i_z} \sum_{i_{a-}} \mathbf{D}_t(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}})$$

where $\mathbf{D}_t$ is the *distribution* of households over $\mathcal{G}_z \times \mathcal{G}_a$

From policy functions to aggregates

- **What we have:** $c^*(z_t, a_{t-1})$ and $a^*(z_t, a_{t-1})$, what *one* household does in every state
- **What the next lecture needs:** aggregates, e.g.

$$A_t^{hh} = \sum_{i_z} \sum_{i_{a-}} \mathbf{D}_t(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}})$$

where $\mathbf{D}_t$ is the *distribution* of households over $\mathcal{G}_z \times \mathcal{G}_a$

- **So we need to simulate the distribution forward.** Two ways:
 1. Monte Carlo: simulate N households, average
 2. Histogram (Young, 2010): iterate on $\mathbf{D}_t$ itself

From policy functions to aggregates

- **What we have:** $c^*(z_t, a_{t-1})$ and $a^*(z_t, a_{t-1})$, what *one* household does in every state
- **What the next lecture needs:** aggregates, e.g.

$$A_t^{hh} = \sum_{i_z} \sum_{i_{a-}} \mathbf{D}_t(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}})$$

where $\mathbf{D}_t$ is the *distribution* of households over $\mathcal{G}_z \times \mathcal{G}_a$

- **So we need to simulate the distribution forward.** Two ways:
 1. Monte Carlo: simulate N households, average
 2. Histogram (Young, 2010): iterate on $\mathbf{D}_t$ itself
- **Notebook:** 05_simulation.ipynb

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N - 1\}$

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N-1\}$
- **Simulation:** forwards in time from $t = 0$, in each period
 1. Draw z_{it} given z_{it-1} from the transition probabilities
 2. Use linear interpolation to evaluate

$$a_{it} = a^*(z_{it}, a_{it-1})$$

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N-1\}$
- **Simulation:** forwards in time from $t = 0$, in each period
 1. Draw z_{it} given z_{it-1} from the transition probabilities
 2. Use linear interpolation to evaluate

$$a_{it} = a^*(z_{it}, a_{it-1})$$

- **Aggregation:** sample averages, $A_t^{hh} = \frac{1}{N} \sum_i a_{it}$
 - Law of large numbers: correct as $N \rightarrow \infty$

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N-1\}$
- **Simulation:** forwards in time from $t = 0$, in each period
 1. Draw z_{it} given z_{it-1} from the transition probabilities
 2. Use linear interpolation to evaluate

$$a_{it} = a^*(z_{it}, a_{it-1})$$

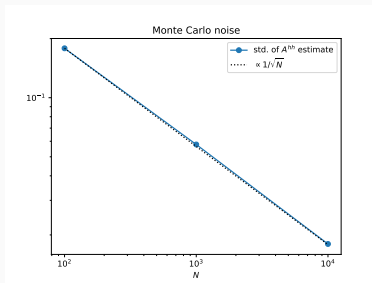
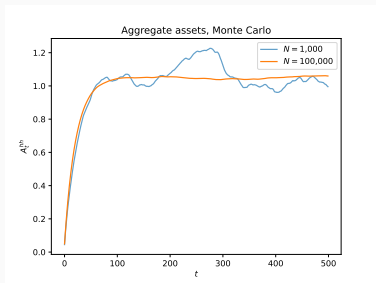
- **Aggregation:** sample averages, $A_t^{hh} = \frac{1}{N} \sum_i a_{it}$
 - Law of large numbers: correct as $N \rightarrow \infty$
- **Pro:** simple to implement, works for any model

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N-1\}$
- **Simulation:** forwards in time from $t = 0$, in each period
 1. Draw z_{it} given z_{it-1} from the transition probabilities
 2. Use linear interpolation to evaluate

$$a_{it} = a^*(z_{it}, a_{it-1})$$

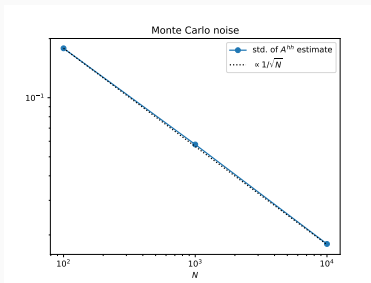
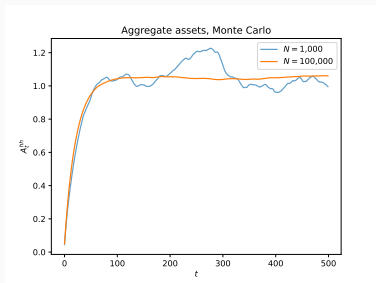
- **Aggregation:** sample averages, $A_t^{hh} = \frac{1}{N} \sum_i a_{it}$
 - Law of large numbers: correct as $N \rightarrow \infty$
- **Pro:** simple to implement, works for any model
- **Con:** computationally costly, and *random*

Monte Carlo noise



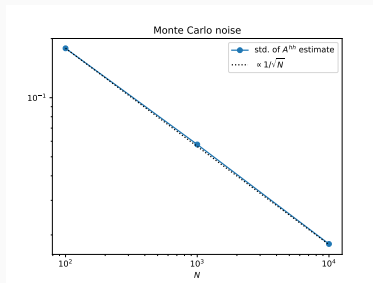
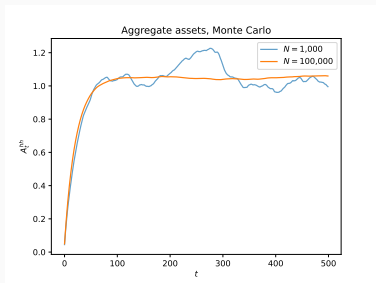
- **Left:** A_t^{hh} never settles, it keeps fluctuating around its stationary level

Monte Carlo noise



- **Left:** A_t^{hh} never settles, it keeps fluctuating around its stationary level
- **Right:** the error falls like $1/\sqrt{N}$: quadrupling N halves it

Monte Carlo noise



- **Left:** A_t^{hh} never settles, it keeps fluctuating around its stationary level
- **Right:** the error falls like $1/\sqrt{N}$: quadrupling N halves it
- **Why this hurts:** next lecture we root-find on asset market clearing, $A^{hh}(r) - K(r) = 0$. A stochastic A^{hh} puts a floor on the achievable tolerance

The distribution as an object

- **Idea:** stop tracking households, track *probability mass*

The distribution as an object

- **Idea:** stop tracking households, track *probability mass*
- **Histogram:** an array $\mathbf{D}$ on $\mathcal{G}_z \times \mathcal{G}_a$, summing to 1
 - $\underline{D}_t$: beginning-of-period, over (z_{t-1}, a_{t-1})
 - D_t : after the productivity draw, over (z_t, a_{t-1})

The distribution as an object

- **Idea:** stop tracking households, track *probability mass*
- **Histogram:** an array $\mathbf{D}$ on $\mathcal{G}_z \times \mathcal{G}_a$, summing to 1
 - $\underline{\mathbf{D}}_t$: beginning-of-period, over (z_{t-1}, a_{t-1})
 - $\mathbf{D}_t$: after the productivity draw, over (z_t, a_{t-1})
- **The productivity transition is exact:**

$$\mathbf{D}_t(z^{i_z}, a^{i_a-}) = \sum_{i_z-} \pi_{i_z-, i_z} \underline{\mathbf{D}}_t(z^{i_z-}, a^{i_a-}) \iff \mathbf{D}_t = \Pi'_z \underline{\mathbf{D}}_t$$

The distribution as an object

- **Idea:** stop tracking households, track *probability mass*
- **Histogram:** an array $\underline{D}$ on $\mathcal{G}_z \times \mathcal{G}_a$, summing to 1
 - $\underline{D}_t$: beginning-of-period, over (z_{t-1}, a_{t-1})
 - $\underline{D}_t$: after the productivity draw, over (z_t, a_{t-1})
- **The productivity transition is exact:**

$$\underline{D}_t(z^{i_z}, a^{i_{a-}}) = \sum_{i_{z-}} \pi_{i_{z-}, i_z} \underline{D}_t(z^{i_{z-}}, a^{i_{a-}}) \iff \underline{D}_t = \Pi'_z \underline{D}_t$$

- **The problem:** all mass at $(z^{i_z}, a^{i_{a-}})$ saves $a^*(z^{i_z}, a^{i_{a-}})$, which falls *between* grid points

The savings lottery

- **Split the mass between the two neighbors.** Find i such that $a^i \leq a^* \leq a^{i+1}$ and set

$$\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i} \in [0, 1]$$

The savings lottery

- **Split the mass between the two neighbors.** Find i such that $a^i \leq a^* \leq a^{i+1}$ and set

$$\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i} \in [0, 1]$$

- **Assign** share ω to a^i and $1 - \omega$ to a^{i+1}
 - The lottery has mean *exactly* a^* : aggregate savings are not distorted by the grid

The savings lottery

- **Split the mass between the two neighbors.** Find i such that $a^i \leq a^* \leq a^{i+1}$ and set

$$\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i} \in [0, 1]$$

- **Assign** share ω to a^i and $1 - \omega$ to a^{i+1}
 - The lottery has mean *exactly* a^* : aggregate savings are not distorted by the grid
- **One full period:**

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

where Λ collects the lottery weights

The savings lottery

- **Split the mass between the two neighbors.** Find i such that $a^i \leq a^* \leq a^{i+1}$ and set

$$\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i} \in [0, 1]$$

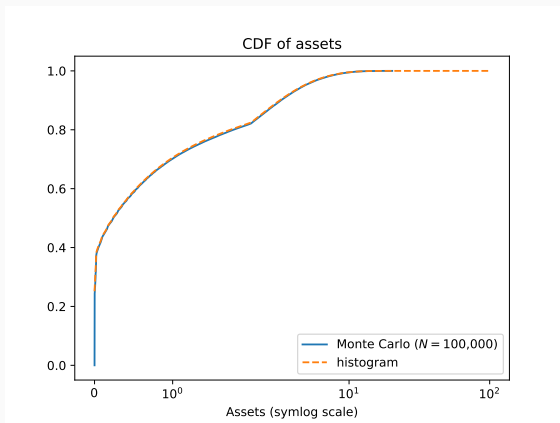
- **Assign** share ω to a^i and $1 - \omega$ to a^{i+1}
 - The lottery has mean *exactly* a^* : aggregate savings are not distorted by the grid
- **One full period:**

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

where Λ collects the lottery weights

- **Both maps are linear and deterministic:** simulation becomes matrix algebra. No draws, no noise

Monte Carlo vs histogram



Same distribution. The histogram is the exact limit the Monte Carlo approaches by brute force, at a fraction of the cost. Its only cost: the distribution lives *on the grid*.

- **Aggregates are sums over the grid:**

$$A^{hh} = \sum_{i_z, i_{a-}} D(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}}), \quad C^{hh} = \sum_{i_z, i_{a-}} D c^*$$

- **Aggregates are sums over the grid:**

$$A^{hh} = \sum_{i_z, i_{a-}} D(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}}), \quad C^{hh} = \sum_{i_z, i_{a-}} D c^*$$

- **Stationary distribution:** iterate the forward step until D stops moving (526 iterations, 0.01 secs)

Summary

The household block in practice

1. Backward step (EGM): iterate

$$\underline{v}_a \rightarrow c^*(z_t, a_{t-1}), a^*(z_t, a_{t-1})$$

until the policy functions converge $\Rightarrow$ solve_hh_backwards

This pair is the household block. Next lecture GEModelTools runs it for us, and r and w become equilibrium objects.

The household block in practice

1. Backward step (EGM): iterate

$$\underline{v}_a \rightarrow c^*(z_t, a_{t-1}), a^*(z_t, a_{t-1})$$

until the policy functions converge $\Rightarrow$ solve_hh_backwards

2. Forward step (histogram): build the lottery from a^* , iterate

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

until the distribution converges $\Rightarrow$ simulate_hh_forwards

This pair is the household block. Next lecture GEModelTools runs it for us, and r and w become equilibrium objects.

The household block in practice

1. **Backward step (EGM):** iterate

$$\underline{v}_a \rightarrow c^*(z_t, a_{t-1}), a^*(z_t, a_{t-1})$$

until the policy functions converge $\Rightarrow$ solve_hh_backwards

2. **Forward step (histogram):** build the lottery from a^* , iterate

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

until the distribution converges $\Rightarrow$ simulate_hh_forwards

3. **Aggregate:** $A^{hh} = \sum D a^*$, $C^{hh} = \sum D c^*$

This pair is the household block. Next lecture GEModelTools runs it for us, and r and w become equilibrium objects.

What you should remember

1. The PIH gives an MPC of about 2%; the data say 40 – 50%

Next: *Stationary equilibrium*

You should:

What you should remember

1. **The PIH gives an MPC of about 2%; the data say 40 – 50%**
2. **Uncertainty plus a borrowing constraint fixes it**

Next: *Stationary equilibrium*

You should:

What you should remember

1. The PIH gives an MPC of about 2%; the data say 40 – 50%
2. Uncertainty plus a borrowing constraint fixes it
3. EGM solves the household problem fast and accurately

Next: *Stationary equilibrium*

You should:

What you should remember

1. **The PIH gives an MPC of about 2%; the data say 40 – 50%**
2. **Uncertainty plus a borrowing constraint fixes it**
3. **EGM solves the household problem fast and accurately**
4. **The histogram method gives a precise distribution, without simulation noise**

Next: *Stationary equilibrium*

You should:

What you should remember

1. **The PIH gives an MPC of about 2%; the data say 40 – 50%**
2. **Uncertainty plus a borrowing constraint fixes it**
3. **EGM solves the household problem fast and accurately**
4. **The histogram method gives a precise distribution, without simulation noise**

Next: *Stationary equilibrium*

You should:

1. Study the code from this lecture

What you should remember

1. **The PIH gives an MPC of about 2%; the data say 40 – 50%**
2. **Uncertainty plus a borrowing constraint fixes it**
3. **EGM solves the household problem fast and accurately**
4. **The histogram method gives a precise distribution, without simulation noise**

Next: *Stationary equilibrium*

You should:

1. Study the code from this lecture
2. Glance at Aiyagari (1994),
»Uninsured Idiosyncratic Risk and Aggregate Saving«

Misc



1. Life-cycle (I)

- **Basically:**

1. Born, working, retired, die
2. Age-varying parameters (esp. income)

1. Life-cycle (I)

- **Basically:**

1. Born, working, retired, die
2. Age-varying parameters (esp. income)

- **Add-ons:**

1. Labor supply, human capital, occupation
 2. Portfolio choice and entrepreneurship
 3. Family formation
 4. Health, mortality
- etc.

1. Life-cycle (I)

- **Basically:**

1. Born, working, retired, die
2. Age-varying parameters (esp. income)

- **Add-ons:**

1. Labor supply, human capital, occupation
 2. Portfolio choice and entrepreneurship
 3. Family formation
 4. Health, mortality
- etc.

- **Good starting example:** »Life-Cycle Consumption and Children: Evidence from a Structural Estimation«, Jørgensen (2017)

1. Life-cycle (II)

Paper: Gourinchas and Parker (2002)

Life-cycle consumption-saving model with retirement

- Young households:
Save for precautionary reasons
(buffer)

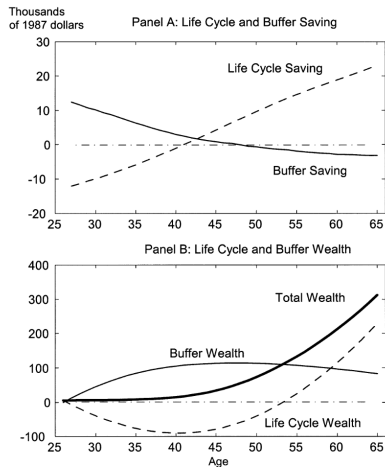


FIGURE 7.—The role of risk in saving and wealth accumulation.

1. Life-cycle (II)

Paper: Gourinchas and Parker (2002)

Life-cycle consumption-saving model with retirement

- Young households:
Save for precautionary reasons (buffer)
- Older households:
Save for retirement (life-cycle)

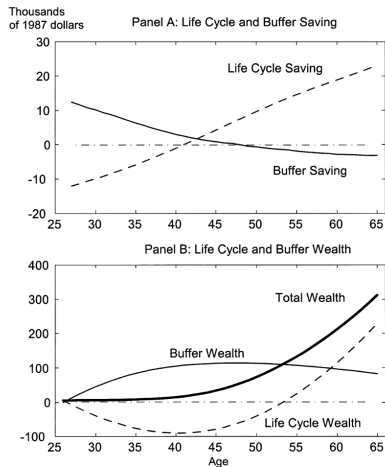
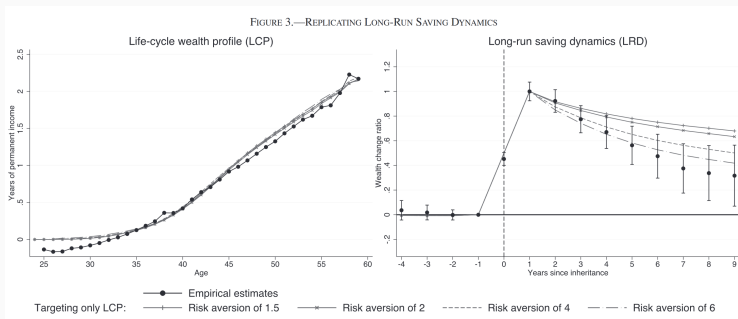


FIGURE 7.—The role of risk in saving and wealth accumulation.

1. Life-cycle (III)

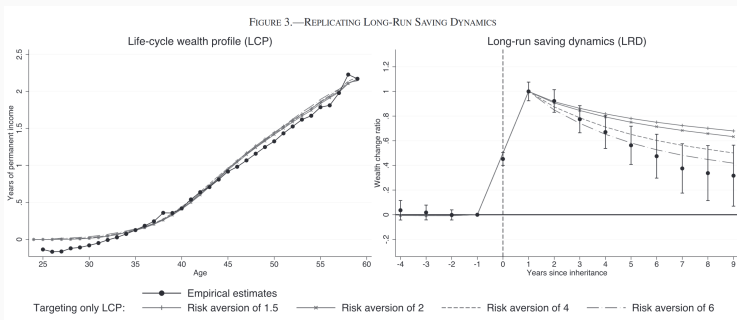
- **Natural experiment:** Wealth depletion after sudden inheritance



Source: Druedahl and Martinello (2022)

1. Life-cycle (III)

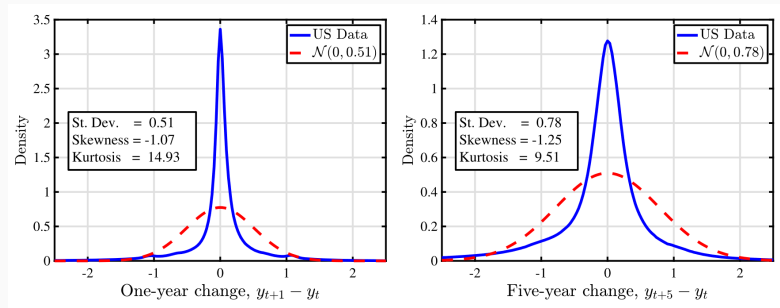
- **Natural experiment:** Wealth depletion after sudden inheritance
- **Results:**
 1. Life-cycle profile of wealth fitted for many levels of risk-aversion (by varying the discount factor)
 2. Fast wealth depletion requires high risk-aversion (or high perceived risk)



Source: Druedahl and Martinello (2022)

2. More realistic income risk (I)

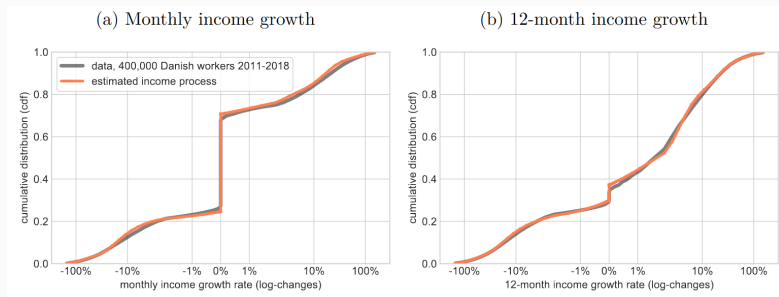
Annual earnings-changes are far from log-normal:



Source: Guvenen et al. (2021)

2. More realistic income risk (II)

Many with zero-growth month-month:



Source: Druedahl et al. (2021)

3. Epstein-Zin

$$\begin{aligned} v_t(z_t, m_t) &= \max_{c_t} \left[(1 - \beta) \cdot c_t^{1-\sigma} + \beta \cdot w_{t+1}^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \\ \text{s.t.} \quad w_{t+1} &\equiv \mathbb{E}_t \left[v_{t+1}(z_{t+1}, m_{t+1})^{1-\rho} \right]^{\frac{1}{1-\rho}} \\ m_{t+1} &= (1 + r)(m_t - c_t) + y_{t+1} \end{aligned}$$

- **Preferences:**

1. Patience: β
2. Intertemporal substitution: σ
3. Risk-aversion: ρ

3. Epstein-Zin

$$\begin{aligned} v_t(z_t, m_t) = & \max_{c_t} \left[(1 - \beta) \cdot c_t^{1-\sigma} + \beta \cdot w_{t+1}^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \\ \text{s.t.} \quad & w_{t+1} \equiv \mathbb{E}_t \left[v_{t+1}(z_{t+1}, m_{t+1})^{1-\rho} \right]^{\frac{1}{1-\rho}} \\ & m_{t+1} = (1 + r)(m_t - c_t) + y_{t+1} \end{aligned}$$

- **Preferences:**

1. Patience: β
2. Intertemporal substitution: σ
3. Risk-aversion: ρ

- **Euler-equation:** $c_t^{-\sigma} = \beta R \cdot \mathbb{E}_t \left[c_{t+1}^{-\sigma} \cdot \left(\frac{w_{t+1}}{v_{t+1}} \right)^{\rho-\sigma} \right]$

1. FOC: $0 = v_t^\sigma \cdot \left[(1 - \beta) \cdot c_t^{-\sigma} - \beta R \cdot w_{t+1}^{\rho-\sigma} \cdot \mathbb{E}_t \left[v_{t+1}^{-\rho} \cdot \frac{\partial v_{t+1}}{\partial m_{t+1}} \right] \right]$
2. Envelope condition: $\frac{\partial v_t(z_t, m_t)}{\partial m_t} = v_t^\sigma \cdot (1 - \beta) \cdot c_t^{-\sigma}$

4. Deep learning

- **Curse of dimensionality:**

1. Many states
2. Many choices
3. Many shocks

4. Deep learning

- **Curse of dimensionality:**

1. Many states
2. Many choices
3. Many shocks

- **Deep (reinforcement) learning:**

1. Approximate value and policy functions with *neural networks*
2. Approximate on simulation sample rather than on grid
3. Automatic differentiation (backpropagation) and GPUs for speed

4. Deep learning

- **Curse of dimensionality:**
 1. Many states
 2. Many choices
 3. Many shocks
- **Deep (reinforcement) learning:**
 1. Approximate value and policy functions with *neural networks*
 2. Approximate on simulation sample rather than on grid
 3. Automatic differentiation (backpropagation) and GPUs for speed
- **Examples:** Maliar and Maliar (2021) and Azinovic and Scheidegger (2022)

4. Deep learning

- **Curse of dimensionality:**
 1. Many states
 2. Many choices
 3. Many shocks
- **Deep (reinforcement) learning:**
 1. Approximate value and policy functions with *neural networks*
 2. Approximate on simulation sample rather than on grid
 3. Automatic differentiation (backpropagation) and GPUs for speed
- **Examples:** Maliar and Maliar (2021) and Azinovic and Scheidegger (2022)
- **Working paper:** Druedahl and Røpke (2025)
Python package: [EconDLSolvers](#)